**Statement of Work (SOW)**
**<u>Feasibility Study: Payload Processing Facility for Nuclear Operations</u>**

**Background:**
NASA anticipates that the number of spacecraft missions involving nuclear material will increase substantially in the coming years. To support this growth, an expansion of Payload Processing Facilities (PPFs) certified for nuclear material handling and storage will be required. These facilities must be capable of accommodating standard mission needs while also meeting rigorous cleanliness levels required by planetary protection protocols and supporting hazardous operations, including spacecraft fueling.

**Purpose:**
NASA requires insight into existing/potential PPFs that will possess the capability to process a Spacecraft with nuclear material by January 1, 2028. This facility shall be located on, or in the vicinity of, Kennedy Space Center (KSC) / Cape Canaveral Space Force Station (CCSFS) and have a means to transport the encapsulated Spacecraft to a launch pad on KSC/CCSFS in accordance with Idaho National Laboratory requirements.

The Contractor shall provide a comprehensive ground-operations concept of flow describing how an offeror-provided PPF would be utilized to process a spacecraft containing nuclear material.

The Contractor shall detail how the proposed facility will generate and deliver all required technical support, analyses, and documentation necessary to meet the requirements of the Department of Energy (DOE), Idaho National Laboratory (INL), and NASA to facilitate attainment of a Hazard Category 3 designation for the processing of nuclear material no later than January 1, 2028.

**Contractor shall:**

1. Develop the Ground Operations Concept of Flow
    a. Prepare a detailed Ground Operations Concept of Flow describing all activities required to receive, process, integrate, and prepare the spacecraft and associated GSE within the PPF, as well as encapsulate into a fairing up to 5-meter in diameter.
    b. Describe operational sequences, required facilities and equipment, safety controls, interfaces, and constraints.
    c. Describe how the facility will meet the following requirements:

        i. ISO Cleanliness ISO class 7
        ii. Planetary Protection Safeguards
        iii. Light Weight Radioisotope Heater Unit (LWRHU) Nuclear Material Handling and Storage
        iv. Spacecraft Processing area, including for LWRHU installation, with minimum dimensions of 21m W x 21m L x 6m H.

v. 5-meter fairing preparation and encapsulation area with minimum dimensions of 15m W x 21m L x 18m H
vi. Spacecraft Monopropellant Fueling Operations
vii. International partners performing some or all of the spacecraft processing

2. Provide the required documentation to DOE/INL and NASA in support of the required safety assessment to facilitate attainment of a Hazard Category 3 designation for processing of nuclear material.
3. Provide PPF specifications, capabilities, and services, if existing.
4. If the offered Payload Processing Facility (PPF) is not yet constructed or not currently in operation, the offeror shall provide NASA with a clear and comprehensive description of the facility's construction process with updates through period of performance. The submission shall include:
   a. Detailed schedule to completion, including major milestones and dependencies.
   b. Clear delineation of schedule margin available throughout the construction and activation timeline.
   c. Identification and analysis of risks associated with completing construction, commissioning, and achieving required operational capability within the period of performance.
   d. Mitigation strategies for each identified risk.
5. Shall obtain TAA with an international partner to allow exchange of information.

**Deliverables:**

1. Provide required facility and hazard type documentation associated with facility in support of obtaining the DOE/INL safety assessment by January 1, 2028.
2. Provide comprehensive description of the facility's construction process to be delivered within 1-week of ATP.
3. Provide a power point presentation by October 1, 2026, summarizing the results of the Ground Operations Concept of Flow.
4. Provide ITAR version of the results to international partners.
5. Support monthly 30-minute telecons to provide status.
6. Provide a ROM price by October 1, 2026, to implement Contractor's Concept of Flow in the processing facility for a 5-month processing starting on January 1, 2028.
7. Provide ROM price by October 1, 2026. for completion of Hazard Category 3 designation for processing of nuclear material no later than January 1, 2028.

The Contractor shall submit all deliverables to the appropriate NASA Contracting Officer, Contract Specialist, Mission Manager and Technical Point of Contact (TBD at Award).